

LECTURE 6: RECURSION

Reading: Supplements for Lecture 6

10 min

Video: Foundations

12 min

Video: A classic example

11 min

Video: Recursive graphics

11 min

Video: Avoiding exponential waste

9 min

Video: Dynamic programming

14 min

Reading: Optional Enrichment on Recursion

10 min

Quiz: Recursion

8h

Programming Assignment: Recursion

8h

Congratulations! You passed!

Grade received 100%

Latest Submission Grade 100%

To pass 80% or higher

Go to next item

Submit your assignment

1. Which of the following are broadly useful approaches for solving problems by combining solutions to smaller subproblems? Mark all that apply.

Correct

Receive grade

To Pass 80% or higher

1 / 1 point

Try again

2. Give the value of Q2(6)

1 public static int Q2(int n) {
2 {
3 if (n <= 0) return 1;
4 return 1 + Q2(n-2) + Q2(n-3);
5 }
6 }
7 }
8 }

Report an issue

Correct

1 / 1 point

3. Give the 10th integer printed by a call to Q3(6).

1 public static void Q3(int n)
2 {
3 if (n <= 0) return;
4 StdOut.println(n);
5 Q3(n-2);
6 Q3(n-3);
7 StdOut.println(n);
8 }

Correct

1 / 1 point

4. Give the number of integers printed by a call to Q4(7).

1 public static void Q4(int n)
2 {
3 if (n <= 0) return;
4 StdOut.println(n);
5 Q4(n-2);
6 Q4(n-3);
7 StdOut.println(n);
8 }

Correct

1 / 1 point

5. Give the value of Q5(8).

1 public static int Q5(int n)
2 {
3 int[] b = new int[n+1];
4 b[0] = 1;
5 for (int i = 2; i <= n; i++)
6 {
7 b[i] = 0;
8 for (int j = 0; j < i; j++)
9 b[i] += b[j];
10 }
11 return b[n];
12 }

Correct

1 / 1 point

https://www.coursera.org/learn/ics-programming-java/exam/uf5Ghrecursion/attempt/redirectToCover-tue

1/1